

AI-ENHANCED INSTITUTIONAL KNOWLEDGE

The Four-Legged Architecture from Protected
Knowledge Vaults to Governed Cognitive Evolution

MFIN PEDAGOGICAL EDITION · CANONICAL NB01–NB22 DIRECTORY

“The strategic unit of artificial intelligence is not the model. It is the governed institutional system through which human judgment, protected knowledge, and generative AI accumulate into institutional memory.”

ALEJANDRO REYNOSO

JUDGE BUSINESS SCHOOL · UNIVERSITY OF CAMBRIDGE · JULY 2026

Abstract

Artificial intelligence is still too often introduced into organizations as a standalone capability: a chatbot, copilot, model endpoint, or collection of isolated pilots. That framing understates both the opportunity and the difficulty. The durable asset of a sophisticated institution is not the generic model it can rent, but the protected body of knowledge through which its people interpret markets, remember decisions, combine evidence, recognize exceptions, and act under uncertainty.

This paper presents an end-to-end architecture for creating *AI-enhanced institutional knowledge*. The pipeline begins with governed source vaults, converts documents and data into canonical objects and typed relations, constructs purpose- and access-conditioned context for a language model, separates generated candidates from authoritative knowledge, and admits only approved increments through reversible write-back. It then shows how the question asked and the route used to navigate the vault condition both the current answer and the topology inherited by future decisions. Classical, learning-based, optimization, quantum-inspired, and quantum methods enter as alternative selectors over admissible evidence paths; they extend the architecture without displacing its governance or its classical baselines.

The architecture is designed to let corporations use generative models while retaining institutional authority over proprietary information. The model does not receive unrestricted access to the vault and does not write directly into canonical memory. A governed interface discloses the minimum eligible context for a declared user and purpose, while a separate gate controls every proposed mutation. This design supports confidentiality, portability, and auditability; it does not by itself guarantee them. Deployment architecture, provider contracts, identity and access management, encryption, retention, residency, and monitoring remain mandatory controls.

This edition is written for the Master of Finance classroom at Judge Business School and must be read in conjunction with the twenty-two Colab notebooks executed in the author's live sessions. Their canonical order and names are those of the NB01–NB22 course directory reproduced in Table A.2. **NB12** is intentionally a shared companion to both the question-induced architecture and the applied *AI-Driven Knowledge Engineering* paper: one explains why purposeful inquiry creates conditional structure; the other shows how that structure is preserved, typed, validated, authorized, and projected back into a working vault. The notebooks demonstrate the mechanisms; this paper supplies the conceptual vocabulary, economic logic, governance boundaries, epistemic limits, and institutional validation design. Together they form a complete teaching pipeline from protected sources to governed, evolving, and reversible institutional knowledge.

Central proposition

The strategic unit of artificial intelligence is not the model. It is the governed institutional system in which human judgment, protected knowledge, and generative AI produce cumulative, AI-enhanced institutional knowledge. Governance determines what the model may see, what the institution may retain, and whether recursive memory compounds insight or error.

Classroom reading contract. This synthesis paper connects *The Governed ExoBrain*, *Question-Induced Knowledge Architecture*, *AI-Driven Knowledge Engineering*, *Navigation-Conditioned Knowledge Evolution*, and *The Quantum Brain*. It is not a substitute for the NB01–NB22 laboratory, and the notebooks are not a substitute for the paper. Students should read the assigned section before class, execute the paired notebook with the author, preserve its audit artifacts, and return to the paper to interpret the result and its limits. Notebook numbers and filenames in the directory are authoritative: every cross-reference in this edition follows that sequence.

Contents

1	Beyond the Chatbot: The Real Object of AI Adoption	6
1.1	The implementation paradox	6
1.2	The scarce asset is not the generic model	6
1.3	From tool adoption to cognitive infrastructure	6
1.4	A collection about implementation, value, and method	7
1.5	The route	8
	Vocabulary card	9
2	The Four Elements of Institutional Intelligence	10
2.1	Complementarity and governed accumulation	10
2.2	Element I: human intelligence	11
2.3	Element II: institutional human knowledge	11
2.4	Element III: generative artificial intelligence	11
2.5	The confidentiality boundary: governed context, not vault access	11
2.6	Element IV: governed AI-enhanced institutional knowledge	12
2.7	The frame: governance	12
	The Canonical Vocabulary	14
3	Why Partial Adoption Fails	15
3.1	Productivity is not transformation	15
3.2	The chatbot trap	15
3.3	The archive trap	15
3.4	The model-maximalist trap: one comparison pass	15
3.5	The automation-first trap	16
3.6	The proof-of-concept trap	16
3.7	The sixth trap: repeating the knowledge-management programme that failed	16
4	From Context to Compounding: Question-Induced Knowledge	18
4.1	The question as an economic and causal object	18
4.2	NB12 as the knowledge-engineering hinge	18
4.3	A fixed model, evolving institutional knowledge	19
4.4	The living encyclopedia	19
4.5	From narrative shock to governed decision state	19
4.6	Contextual compounding and its limit	20
4.7	The danger of epistemic self-consumption	20
5	Navigation Is a Policy, Not Plumbing	21
5.1	The graph the model has is not the graph it sees	21
5.2	Different routes embody different styles of inquiry	21
5.3	The recursive effect: a worked NB15 fork	21
5.4	Contradiction as an institutional asset	22
5.5	Explainable access	22
5.6	Plural navigation as checks and balances	22
5.7	From paths to a context portfolio	22
5.8	Access-conditioned navigation	23

6	Governance in Operation	25
6.1	Two coupled lanes	25
6.2	Integrity, determinism, traceability, and auditability	25
6.3	Layered validation and orphan control	25
6.4	Risk-tiered autonomy	25
	What the notebooks cannot show you	26
6.5	The audit record is also a discovery liability	26
6.6	Reviewer capacity is the binding constraint	26
6.7	Knowledge depreciation: a mechanism, not a symbol	26
6.8	Mapping to model-risk management	27
6.9	Governance creates option value	27
7	The Economics of a Low-Cost, Modular, Model-Agnostic Architecture	29
7.1	The adoption objective	29
7.2	Modularity	29
7.3	Scalability through reusable objects	29
7.4	The cost curve	29
7.5	Model agnosticism as bargaining power	29
8	The Architecture in Finance	31
8.1	Why finance is the proving ground	31
8.2	A running example: the governed investment committee	31
8.3	The finance decision cycle	31
8.4	Representative applications	31
8.5	Separation of duties in an agentic system	31
8.6	The competitive advantage	31
9	Implementation: From Pilot to Cognitive Operating System	33
9.1	The governing rule: build the substrate before autonomy	33
9.2	Stage 0: define the value case and constitutional perimeter	33
9.3	Stage 1: governed sources	33
9.3.1	Entity resolution	33
9.3.2	Corpus reality	33
9.4	Stage 2: canonical objects and graph	34
9.5	Stage 3: inquiry and navigation	34
9.6	Stage 4: context portfolios	34
9.7	Stage 5: deliberation, governed write-back, and reversal	35
9.8	The minimum production stack	35
9.9	Organization and ownership	35
10	Institutional Validation Protocol	37
10.1	The reader's experiment	37
10.2	The five-arm comparison	37
10.3	Controls, measures, and sequence	37
10.4	Knowledge contribution by ablation	38
10.5	Portability and user-relative topology audits	38
10.6	The claim ladder for advanced methods	39
10.7	Pre-registered refutation conditions	39

11 Leadership Agenda and Conclusion	41
11.1 What leaders must decide	41
11.2 Ten commitments for institutional adoption	41
11.3 The close: intellectual transition, economics, and responsibility	41
A Course sequence, sessions, and notebook ladder	42
A.1 Paper-section-to-session map	42
A.2 Full NB01–NB22 ladder	43
A.3 One cohort, one order	45
B Minimum Canonical Contract	46
B.1 Canonical knowledge object	46
B.2 Relationship object	46
B.3 Navigation record	46
B.4 Reversal record	47
B.5 The canonical discipline	47
C Graded Assignment: Pre-register an Institutional Test	48
C.1 Brief	48
C.2 Required submission	48
C.3 Submission template	49

1 Beyond the Chatbot: The Real Object of AI Adoption

1.1 The implementation paradox

Generative AI has made intelligence appear inexpensive. A model can draft a memorandum, summarize a filing, compare companies, generate code, or propose a strategy in seconds. This changes the marginal cost of producing plausible cognitive output. It does not, by itself, change the institution's capacity to know.

An answer can be excellent and still vanish at the end of a conversation. It can be preserved and lose its provenance. It can be retrieved later and be mistaken for verified evidence. It can be repeated by several agents and acquire the appearance of consensus even though every repetition descends from one machine-generated hypothesis. It can be operationally useful while remaining institutionally illegible.

This is why early enterprise pilots often combine enthusiastic users, impressive demonstrations, and weak organizational learning. The model performs; the institution does not necessarily learn. Individual tasks accelerate while knowledge remains fragmented across chats, documents, inboxes, personal judgment, and undocumented exceptions.

The institutional question

How can an organization combine human judgment, proprietary knowledge, generative models, and an evolving external memory so that cognition becomes cumulative, governable, portable, and economically valuable?

1.2 The scarce asset is not the generic model

Frontier models are powerful, but their general capabilities are available to many firms. Access can be purchased, providers can change, prices can fall, and competitors can license the same capability. The differentiating assets are the accumulated pattern recognition of professionals, rationales behind past transactions, interpretations of constraints, histories of counterparties, and the difference between formal policy and practical execution. These are a distributed institutional theory of the world, not merely a collection of files.

Much of this knowledge is only partly explicit. Some lives in authoritative documents; some in models, spreadsheets, committee minutes, and the memory of a banker who recalls why a superficially attractive transaction failed fifteen years earlier. The challenge is to externalize enough of this knowledge to make it reusable without pretending that representation eliminates judgment.

1.3 From tool adoption to cognitive infrastructure

Five companion works widen the architectural perimeter. The ExoBrain establishes the constitutional substrate: protected sources, candidate knowledge, governance records, human decisions, canonical objects, relationships, registries, and reversible write-back. Question-Induced Knowledge Architecture makes purposeful inquiry a source of reusable experience without changing model weights. AI-Driven Knowledge Engineering supplies the operational bridge from a preserved document vault to typed corporate entities, computable shocks, recommendation and decision objects, temporal supersession, and a rebuildable graph projection [19]. Navigation-Conditioned Knowledge Evolution shows that stored knowledge is not the same as knowledge made active: the route through memory shapes the answer and the future graph. The Quantum Brain asks whether richer distributions over reasoning paths improve deliberation while preserving strong classical baselines and a strict claim ladder.



Figure 1. The architectural chain: an answer becomes institutional intelligence only after grounding, governed memory, and cumulative reuse.

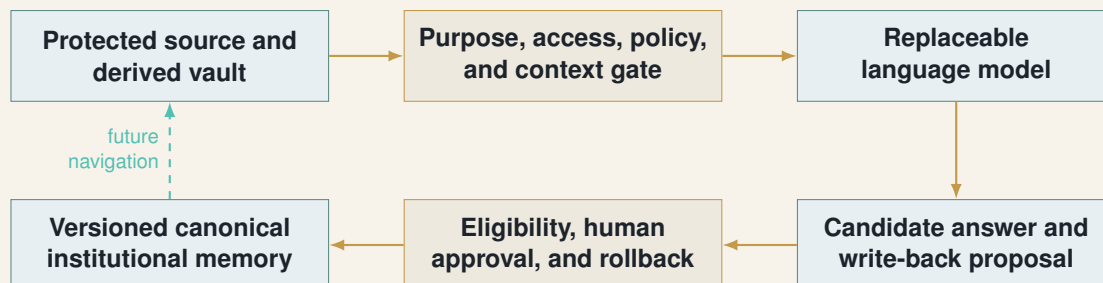


Figure 2. The governed context boundary. There is no direct model-to-vault write path: protected evidence is disclosed through a role- and purpose-conditioned gate, and generated output returns as a non-authoritative proposal.

The final object is not a better prompt. It is a cognitive operating system: an architecture that manages knowledge identity, permissions, navigation, reasoning, action, and history in a manner analogous to the way a conventional operating system manages files, processes, memory, and devices.

1.4 A collection about implementation, value, and method

This collection is about implementation economics, not speculative autonomy. It asks how declining computational costs can become compounding institutional capacity. It has no proprietary institutional corpus, operating process, or deployment budget, and therefore does not present synthetic results as estimates of production value. It isolates mechanisms in reproducible laboratories, states observations that would count against the architecture, and supplies a protocol that institutions with private data can execute.

The pedagogical sequence moves from constitution to computation. Students first learn what can enter the vault and with what authority; only then do they connect a model through a governed context interface, permit recursive write-back, compare navigation policies, construct evidence portfolios, and test advanced selection methods. This order is substantive. Beginning with an impressive model answer would conceal the institutional problem the course is designed to solve.

The laboratory

You will not be given a corpus; you will be given a laboratory. It is synthetic because navigation's effect is visible only when the graph, model, prompt, question, governance criteria, and context budget are frozen. If those quantities move together, an answer difference cannot be attributed to the navigation policy. The synthetic design is therefore an identification device, not a substitute for institutional evidence.

Computer science supplies formal objects, retrieval methods, graphs, agents, optimization, and reproducible pipelines. Economics and finance supply the correct objective: scarce attention, transaction costs, incentives, option value, asymmetric information, model risk, irreversibility, and accountable capital allocation. A system can be technically elegant and economically irrelevant; it can be commercially ambitious and epistemically unsafe.

1.5 The route

The classroom build follows a six-stage spine. This is the route to remember; the full notebook-by-notebook ladder, including the concept taught and the failure created by skipping a rung, is Table A.2 in Appendix A.2. The paper-section-to-session map is Table A.1 in Appendix A.1.

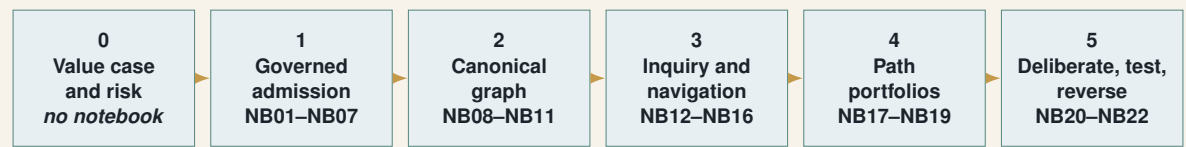


Figure 3. The six-stage teaching and implementation spine, adapted from the paper’s original staged path. Each stage is useful on its own and creates the preconditions for the next.

Stage 0 has no notebook. Stages 1–5 establish, in order, governed admission, the canonical graph, navigation-conditioned inquiry, evidence-path portfolios, and governed deliberation, benchmarking, and reversal. **NB22** closes the loop by poisoning an approved edge, tracing its dependency cone, and proving complete restoration.

Vocabulary card

epistemic_class	The declared basis of knowing: observed; asserted-by-source; derived-by-model; inferred-under-assumption; or contested. It does not encode confidence or governance approval.
Candidate vs. canonical	A candidate is machine- or human-proposed content with identity and provenance but no authority. A canonical object is the institution's stable, approved, versioned representation; canonical does not mean eternally true.
Stored vs. cognitive topology	Stored topology is the complete permissioned graph. Cognitive topology is the effective subgraph experienced under a question, navigation policy, budget, stopping rule, and user access profile.
Governance status	Lifecycle state—candidate, eligible, quarantined, rejected, approved, canonical, superseded, archived, or erased under mandate.
Navigation record	The reproducible account of seeds, visited and retained nodes, rejected branches, policy, budget, access, stopping reason, and context hash.
Write-back	A versioned proposal to add, qualify, link, supersede, or deprecate knowledge after a reasoning episode.
Governed context boundary	The interface that releases only purpose-, user-, and policy-eligible evidence to a model and prevents direct model mutation of canonical memory.
Supersession	Replacement that preserves lineage: the new object becomes active while the earlier object remains reconstructible.
Quarantine	A non-authoritative holding state for content that is plausible or structurally valid but cannot yet pass the required gate.
Integrity	Whether admitted bytes or declared objects changed unexpectedly.
Determinism	Whether the same declared inputs reproduce the same registry, object identities, and fingerprints.
Traceability	Whether an object can be followed to sources, transformations, rules or models, policy, and reviewer.
Auditability	Whether an independent party can reconstruct the permitted evidence state, reasoning route, decision, and resulting mutation.

Keep this card

These terms are needed in **NB01**; they reappear in navigation governance, institutional validation, model-risk review, audit, and discovery. The software will change faster than the distinctions.

2 The Four Elements of Institutional Intelligence

2.1 Complementarity and governed accumulation

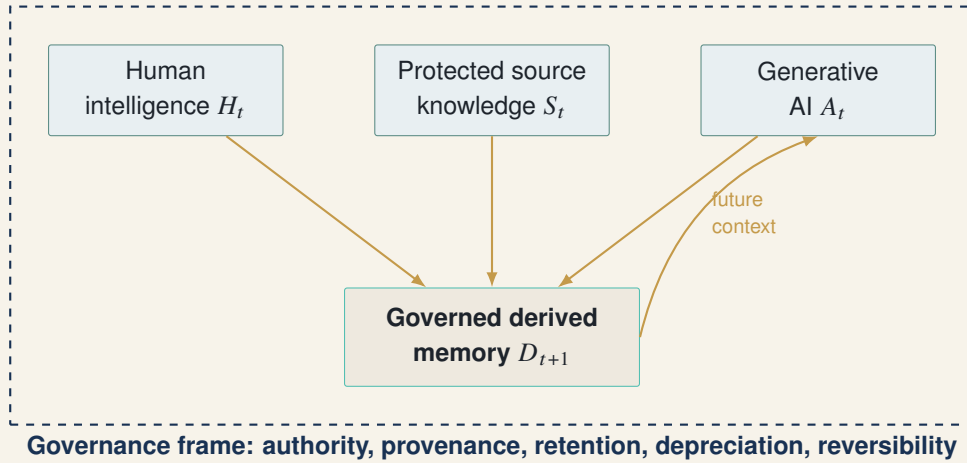
The four-legged title is shorthand for four indispensable elements, not four symmetrical inputs. A stronger model does not repair missing provenance. A larger archive does not supply judgment. Expert talent does not automatically scale. A graph does not decide how it should be traversed.

Human intelligence H_t , protected institutional source knowledge S_t , and generative capacity A_t are current inputs. Governed AI-enhanced institutional knowledge D_t is the inherited stock produced when selected reasoning episodes survive validation, write-back, and later maintenance. Governance is not a fifth substitutable factor. It defines the admissible constraint set.

Mnemonic: four complements inside a governance frame

$$I_t = g_t \cdot P(H_t, S_t, A_t, D_t)$$

This is a teaching mnemonic, not an estimable production function. Governance multiplies rather than adds: at $g_t = 0$, no amount of A_t rescues the product. The notation emphasizes complementarity and a binding constitutional frame; the update equations below carry the operational content.



The accumulation pair is:

$$D_{t+1} = (D_t \ominus L_t) \oplus \Delta D_t^{\text{approved}}, \quad K_t^+ = S_t \oplus D_t.$$

Here L_t is knowledge depreciation and $\oplus$ denotes provenance-preserving addition or **supersession** (Vocabulary Card, p. 9), never silent overwrite. The operator $\ominus$ means demotion from active use to an archival state, not removal. The source stock remains distinguishable from validated extensions.

For operational inspection, the corresponding engineered state can be written as

$$\mathcal{K}_t = (V_t, E_t, \Theta_t, \Pi_t, \Gamma_t),$$

where V_t and E_t are versioned nodes and typed relations, Θ_t contains properties and epistemic states, Π_t contains provenance, and Γ_t contains policy and validation constraints. This representation does not replace K_t^+ ; it makes the accumulated stock testable. A graph is one artifact of knowledge engineering, while the engineering discipline determines how its contents are extracted, resolved, validated, authorized, superseded, and rebuilt.

2.2 Element I: human intelligence

Human intelligence supplies purpose, judgment, imagination, responsibility, social interpretation, and authority. “Human in the loop” is too weak if it means a fatigued reviewer clicking approval. The object model must identify who frames the question, validates the evidence, accepts the interpretation, and authorizes action.

Human intelligence is heterogeneous. A junior analyst may find a data inconsistency that a committee overlooks; a veteran banker may recognize an analogy absent from the record; a lawyer may identify a jurisdictional constraint that invalidates a financially sound recommendation. The system should preserve this plurality rather than convert hierarchy into epistemic certainty.

2.3 Element II: institutional human knowledge

Institutional human knowledge is the proprietary source memory created or validated through institutional activity: documents, data, concepts, relationships, practices, decisions, rationales, exceptions, outcomes, and unresolved disagreements. Three distinctions are essential:

1. Documents are not yet knowledge architecture. A file system may preserve artifacts without exposing which claims are active or what superseded them.
2. Presence is not authority. A file may exist because it was downloaded, drafted, or generated; corpus admission must be explicit.
3. Explicit knowledge is not the whole institution. Human annotation, challenge, and contextualization remain necessary.

Canonical objects (Vocabulary Card, p. 9) create stable intellectual handles without severing content from its source. They do not freeze tacit knowledge into brittle schemas; they make contestable claims, assumptions, and decisions addressable. Each object carries a separate **epistemic_class** and **governance status** (Vocabulary Card, p. 9); how a proposition is known is not the same as whether the institution has approved its current use.

2.4 Element III: generative artificial intelligence

Generative AI supplies a broad prior over language, concepts, and reasoning forms. It can translate, summarize, compare, classify, propose relations, construct hypotheses, simulate objections, write code, and coordinate tools. Its strength is generality; its limitation is the same. It may know the grammar of an acquisition memorandum without knowing why this institution rejects a superficially attractive counterparty.

The model is a reasoning engine, not the institution’s canonical memory or final authority. Parameters and vendors can change independently of the institution. The model is therefore best positioned behind a stable interface to governed knowledge.

2.5 The confidentiality boundary: governed context, not vault access

The phrase “connect the vault to the LLM” is convenient but imprecise. The model should not browse the repository as an unconstrained user, ingest the entire corpus into a prompt, or acquire permission to alter stored knowledge. The **governed context boundary** (Vocabulary Card, p. 9) receives a declared question, user, purpose, access profile, and budget; resolves eligible sources and relations; applies classification, need-to-know, temporal, jurisdictional, and residency rules; and releases only the minimum context needed for the episode.

The return path is deliberately separate. Model output enters a candidate store with its prompt, model, context hash, provenance, and access lineage. It cannot become canonical memory until procedural checks and the required human authority approve an exact write-back proposal. The resulting architecture is therefore asymmetric:

vault $\rightarrow$ governed context $\rightarrow$ model $\rightarrow$ candidate $\rightarrow$ governance $\rightarrow$ canonical memory.

This arrangement allows a corporation to keep knowledge external to model weights and portable across approved models. It also limits the blast radius of a model or provider failure. It is not, however, a claim that confidentiality arises automatically. The institution must still choose an acceptable deployment pattern—for example an approved enterprise endpoint, a private tenant, or a locally controlled model—and enforce identity, encryption, logging, prompt and output retention, provider-training exclusions, data residency, key management, incident response, and contractual audit rights.

2.6 Element IV: governed AI-enhanced institutional knowledge

Governed AI-enhanced institutional knowledge is not simply AI with more context. It is the evolved external memory produced by repeated interactions among human questions, generative models, graph-navigation policies, and authorized **write-back** (Vocabulary Card, p. 9). It remains owned by the institution, external to model weights, portable across approved models, attributable to sources, and subject to human authority.

Let G_t be the versioned graph induced by $K_t^+ = S_t \oplus D_t$. For question q_t , policy π_t , budget B_t , and constraints Ω_t governing user, purpose, access, classification, time, residency, retention, and stopping:

$$\begin{aligned} C_t &= N(G_t, q_t; \pi_t, B_t, \Omega_t), \\ \widetilde{\Delta D}_t &= F_{\theta_t}(q_t, C_t, \Pi_t), \\ D_{t+1} &= (D_t \ominus L_t) \oplus \Gamma_t(\widetilde{\Delta D}_t; S_t, D_t, C_t), \\ K_{t+1}^+ &= S_{t+1} \oplus D_{t+1}. \end{aligned}$$

The navigation operator creates the model’s **cognitive topology** (Vocabulary Card, p. 9): the effective subgraph made available by the stored graph, question, policy, budget, stopping rule, and access profile. Navigation therefore enters the architecture at the same point as context construction, not later as a retrieval implementation detail.

The governance operator Γ_t may approve, qualify, supersede, **quarantine** (Vocabulary Card, p. 9), or reject the proposal. It must also verify that the proposed object and every induced relation can lawfully exist in the access lattice of the destination state. Rejection leaves authoritative state unchanged. Because an approved increment changes the graph available to later navigation, the route through memory affects the present answer and the future institutional knowledge against which later questions are evaluated.

2.7 The frame: governance

Governance separates generation from admission, procedural eligibility from human approval, approval from execution, and canonical representation from timeless truth.

Constitutional rule

Candidate knowledge (Vocabulary Card, p. 9) is not canonical knowledge. Machine generation is not admission. Machine validation is not human approval. Human approval is not execution. Canonical representation is not absolute truth.

The Canonical Vocabulary

Why policy documents are insufficient

Governance must be encoded in object types, states, permissions, gates, and records. If generated and sourced content occupy the same untyped field, no later policy can reliably restore their difference. If an approval is not attached to a versioned proposal, its scope cannot be reconstructed.

The canonical taxonomy

Table 3. Canonical taxonomy: the minimum distinctions required for governed accumulation.

Object	Purpose	Authority class
Source artifact	Preserved record of what entered the corpus	Source evidence
Candidate knowledge object	Proposed statement, hypothesis, relation, or representation	Provisional
Governance record	Evidence of eligibility checks and policy application	Procedural
Human decision	Approval, rejection, qualification, or supersession	Human authority
Canonical knowledge object	Stable representation of a claim, definition, decision, assumption, or outcome	Governed representation
Relationship object	Typed, attributable edge between canonical objects	Governed assertion
Write-back proposal	Exact proposed mutation, scope, effect, and rollback	Not executed
Registry	Versioned identities, fingerprints, status, and lineage	Authoritative index
Navigation record	Policy-conditioned route, retained context, budget, access, and stop	Procedural evidence

Table 3 disambiguates three phrases that are often collapsed: *candidate knowledge* is proposed content; a *write-back proposal* is the exact mutation package submitted to authority; an *approved increment* is the mutation that has passed the applicable gate and may alter canonical state.

Numbering note. Table numbers are retained across editions so that notebook instructions and classroom references remain stable. The move of the canonical taxonomy forward therefore places Table 3 before Tables 1 and 2, which appear in the failure and navigation sections where they are taught.

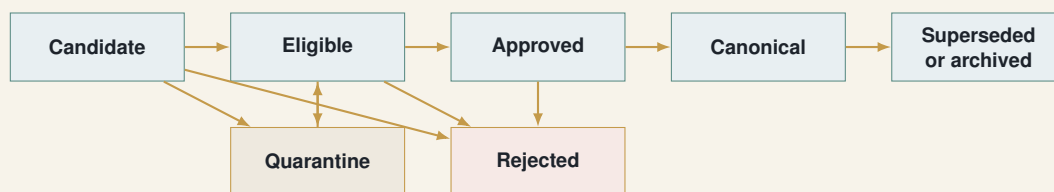


Figure 4. Lifecycle of a governed knowledge proposal. Quarantine preserves a candidate for later evidence; rejection preserves its record but never changes canonical state; supersession and archival preserve lineage.

3 Why Partial Adoption Fails

3.1 Productivity is not transformation

Partial systems can be useful. A model can improve drafting; a document index can accelerate search; a graph can clarify relationships; an approval workflow can reduce a narrow operational risk. Productivity reduces the cost of a present task. Institutional transformation changes the capacity to perform future tasks, transfer knowledge across people, adapt to shocks, and preserve accountability.

Table 1. Why partial adoption does not by itself produce cumulative institutional intelligence.

Missing component	What remains possible	Structural failure
Human intelligence	Automated production and pattern synthesis	No legitimate purpose, accountable judgment, or owned consequence.
Institutional knowledge	Generic assistance and public-domain synthesis	Outputs remain generic; local exceptions and precedent disappear.
Generative AI	A well-governed archive and conventional analytics	Knowledge remains expensive to interpret and recombine.
Governed derived memory	Current analysis and one-off model use	Reasoning does not accumulate as portable, contestable experience.
Governance frame	Rapid experimentation and frictionless write-back	Authority collapses into fluency and recursive error becomes possible.

3.2 The chatbot trap

The chatbot is optimized for the conversation. It may cite sources and produce strong work, but the reasoning episode usually does not become a versioned, reusable institutional object. A transcript preserves words; it does not naturally preserve the chain by which a claim became operational.

3.3 The archive trap

An archive grows by accumulation. A knowledge architecture grows by explicit identity, relation, status, authority, and lifecycle. Search can find a document containing a recommendation; it cannot necessarily answer which assumptions supported it, which outcomes later challenged it, or which current decisions depend on it.

3.4 The model-maximalist trap: one comparison pass

More capable models and larger context windows are valuable. They do not decide which documents deserve authority, how conflicting versions should be treated, or which reasoning output should survive the session. The relevant alternatives are complementary implementation strategies, not rival slogans.

As Table 4 shows, fine-tuning changes model behavior, long context enlarges the working set, RAG grounds a session, GraphRAG supplies corpus-derived structure, and governed write-back accumulates institutional experience outside model weights. A production architecture may use all five, but it must not confuse any one of them with the complete institutional memory system.

Table 4. Fine-tuning, long context, RAG, GraphRAG, and governed external memory solve different problems.

Strategy	Best use	Knowledge update	Provenance/reversal	Strategic limitation
Long context	Bounded multi-document synthesis	Replace or append prompt input	Usually document-level	More room for cognition; no durable lifecycle.
Fine-tuning	Stable behavior or specialized pattern	New training cycle	Weak at episode level	Opacity, staleness, access control, and lock-in.
Static RAG	Current factual grounding	Re-index sources	Strong source citation	Prior reasoning normally remains unstructured.
Static GraphRAG	Relational and corpus-wide sensemaking	Rebuild graph and summaries	Graph lineage can be added	Structure is mainly corpus-induced.
Governed ExoBrain	Persistent institutional cognition	Versioned objects and controlled write-back	Object-level provenance, status, and reversal	Higher architectural and operating discipline.

3.5 The automation-first trap

An agent that operates over weakly governed memory can execute the same mistake repeatedly. The risk becomes circular when the agent modifies the knowledge base that later justifies its own behavior. The safe order is:

Governance → Knowledge architecture → Reasoning → Autonomy.

This does not postpone value until perfection. Low-risk, reversible actions can be automated early; high-consequence knowledge admission and execution retain authority gates.

3.6 The proof-of-concept trap

A production system carries operating cost, review burden, epistemic risk, and lock-in. These quantities do not share a natural unit and should not be hidden inside a decorative objective function. A pilot that produces an impressive memorandum but cannot reproduce its evidence state, transfer learning to another model, or measure whether stored insights improve later decisions has demonstrated a task, not an institution.

3.7 The sixth trap: repeating the knowledge-management programme that failed

Many taxonomy, master-data, and enterprise knowledge-management initiatives failed for three concrete reasons. First, contribution cost fell on non-beneficiaries: experts were asked to stop producing client work to create a public good for somebody else. Second, hand-maintained taxonomies rotted as business language, products, counterparties, and ownership changed. Third, nothing reliably consumed the output inside daily work, so contribution never became a self-reinforcing operating loop.

Generative extraction changes that cost structure. Models can propose object boundaries, metadata, entity matches, relations, summaries, contradictions, and candidate supersession links at machine cost. This is the actual discontinuity. It does not make machine output authoritative; it makes the production of *reviewable candidates* cheap enough that human effort can move from clerical contribution to judgment. NB01–NB03 embody precisely this division: machine-cheap proposal, deterministic eligibility, and human authority.

The old programme asked people to manufacture the whole representation. The new architecture asks them to govern a machine-generated candidate flow. That flow still creates queueing, incentive, and

quality problems. Section 6.6's reviewer-capacity constraint is the *same burden reappearing at approval instead of authoring*. Extraction may be cheap while review, exception handling, and depreciation remain scarce. Consequently, the architecture succeeds only if contribution enters normal work, accepted objects are reusable, reviewers receive capacity and status, and the institution is willing to discard machine-generated structure that does not earn its maintenance cost.

The anti-fragmentation principle

Every local AI use case should consume governed institutional knowledge, contribute governed candidate knowledge, or both. Otherwise adoption scales activity faster than it scales intelligence.

4 From Context to Compounding: Question-Induced Knowledge

Classroom pairing. Read this section before **NB12** (*Question-Induced Knowledge Architecture*) and return to it during **NB14–NB15**. In the live sessions, the author holds the model and initial vault fixed, changes the question or navigation policy, and then opens the write-back gate so that students can distinguish a one-period answer difference from a persistent change in institutional memory.

NB12 has two deliberately complementary readings. This section explains the causal claim: a question changes relevance, induced relations, and future memory. The companion *AI-Driven Knowledge Engineering* paper explains the operational claim: those induced relations become institutional knowledge only after source preservation, entity resolution, ontology control, deterministic transformation, layered validation, authorization, and versioned projection [19]. The shared notebook is therefore a hinge, not a duplication.

4.1 The question as an economic and causal object

Data does not determine relevance by itself. A purposeful question supplies an objective, comparison set, horizon, and implicit loss function. The same company universe presents a different relational surface when the question concerns merger compatibility, rate vulnerability, technological optionality, regulatory exposure, or liquidity under stress.

The question is therefore a first-class object. It records who asked, why the question mattered, which criteria it introduced, what evidence was retrieved, what assumptions were applied, what answer was proposed, and what write-back survived. A transcript preserves words; a reasoning episode preserves an institutional unit of experience.

4.2 NB12 as the knowledge-engineering hinge

The notebook begins with preservation rather than inference. It maintains three logically separate workspaces:

1. an **original vault**, treated as authoritative input and read only;
2. a **security copy**, verified against a content-hash manifest; and
3. a **destination vault**, where extracted objects, derived metrics, candidate relations, and approved projections are built.

The three-vault arrangement is a laboratory implementation of the source, recovery, and engineered-state distinction. A production system may implement the same contract with immutable snapshots, versioned object storage, and a transactional branch rather than three visible directories. The invariant is the same: enrichment must not silently rewrite the evidence from which it was derived, and the starting state must remain reconstructible.

The question then operates over a controlled corporate ontology rather than untyped prose. In the M&A laboratory, the principal objects are *Company*, *Environment Shock*, *Recommendation*, *Committee*, *Decision*, *Event*, and *Reassessment*. Relations distinguish buyer and target roles, stress-testing, review, decision production, outlook updates, event triggers, reassessment, and supersession. The model may propose instances of this ontology; it may not redefine the ontology or authorize the resulting state transition at run time.

The operational-orphan test

A document can be accurate and an extracted claim plausible while both remain operationally orphaned. If the referenced entities, source evidence, question, scenario, decision, and authority do not resolve to registered objects, the system cannot determine what depends on the claim, what must change if it is corrected, or whether a later agent may use it.

4.3 A fixed model, evolving institutional knowledge

The model need not learn in its parameters for the institution to learn. Hold the model fixed while the governed external state evolves. Later questions can then inherit validated relations, contradictions, outcomes, and decision rationales from earlier episodes. This is contextual compounding, not fine-tuning.

4.4 The living encyclopedia

The M&A laboratory begins with a synthetic universe and builds:

1. initial merger hypotheses based on strategic and financial compatibility;
2. macro-shock exposures and explicit revision links;
3. innovation-created opportunities connecting previously distant firms;
4. final recommendations conditioned on the history of questions, shocks, and revisions.

The laboratory demonstrates that the pipeline executes end to end and that governed write-back changes later trajectories. It can establish mechanism, identity, provenance, path dependence, and the reproducibility of a controlled counterfactual.

4.5 From narrative shock to governed decision state

NB12 makes one especially important move from language to computable knowledge. For company i , it represents a transparent sensitivity vector

$$\beta_i = (\beta_{i,r}, \beta_{i,c}, \beta_{i,g}, \beta_{i,a}, \beta_{i,d}, \beta_{i,q}),$$

covering rates, credit, growth, AI productivity, digital substitution, and regulated transition. A scenario s is represented over the same dimensions by z_s , and the illustrative exposure is

$$I_{i,s} = \beta_i^\top z_s.$$

The quantity is a reproducible feature transformation, not a calibrated valuation model. Its pedagogical function is to show how a sentence such as “credit conditions deteriorate” becomes a versioned shock node, a visible calculation, an attributable edge property, and a traversable dependency. The language model explains and compares; deterministic code calculates and resolves.

The worked knowledge cycle then separates proposal from authority and change from overwrite:

An authorized decision is therefore a knowledge-state operation. It *changes active status, creates machine-visible obligations, changes later retrieval, records accountable authority, and establishes the temporal reference point* from which reassessment must proceed. A fluent recommendation can do none of these merely by being written.

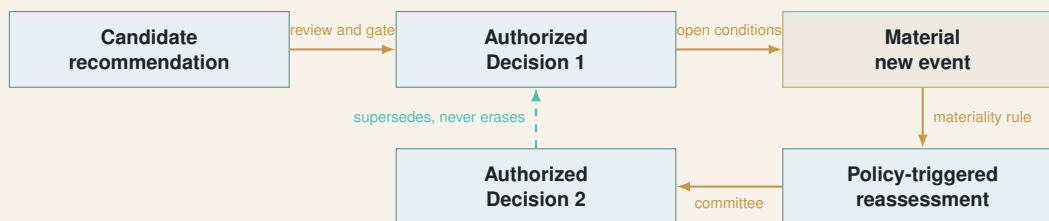


Figure 5. NB12's two-round knowledge cycle. New evidence reopens a governed decision and preserves the causal sequence through explicit supersession.

What the M&A laboratory cannot establish

The corpus and held-out cases share a synthetic generator. The experiment therefore cannot establish institutional value, real-world generalization, or economic advantage. A system can perform well by exploiting regularities of the generator that created both its memory and its test cases. Production value requires proprietary evidence, genuinely held-out decisions, observed outcomes or blinded expert assessment, and review-cost accounting. This is a boundary of the claim, not a footnote.

Operative sentence. The generator that produced the 1,000 companies also produced the held-out cases, so uplift shows that the pipeline recovers inserted structure, not that the structure exists in the world.

4.6 Contextual compounding and its limit

Let P_t denote held-out performance after t governed episodes. The architecture predicts that selected experience can raise future performance, but not monotonically and not without maintenance. Accumulation can saturate, contradict itself, or become stale. The mechanism for depreciation is specified in Section 6.7; the contribution of an object is tested through $KC(k)$ in §10.4.

4.7 The danger of epistemic self-consumption

Once generated structure influences future retrieval, yesterday's reasoning helps determine tomorrow's evidence. Ten apparent confirmations may descend from one unsupported episode. Safe recursion therefore requires source, question, transformation, epistemic status, navigation provenance, and authorization to remain recoverable. The system must distinguish independent support from inherited repetition.

5 Navigation Is a Policy, Not Plumbing

Classroom pairing. Read §5.1–§5.6 before **NB13–NB16**, and §5.7 with the claim ladder before **NB17–NB19**. The objective is not to memorize retrieval algorithms. It is to learn that an apparently technical choice about traversal changes the evidence made cognitively available and, after governed write-back, can alter the state inherited by the next decision.

5.1 The graph the model has is not the graph it sees

A mature vault contains more entities, claims, decisions, shocks, outcomes, and contradictions than any context window can hold. Navigation constructs the active context:

$$C_t = N(G_t, q_t; \pi_t, B_t, \Omega_t).$$

The **stored topology** (Vocabulary Card, p. 9) G_t is not the cognitive topology. Cognitive topology is the effective knowledge space experienced under a particular policy, query, budget, stopping rule, and permission regime. A claim can be physically present and functionally forgotten because approved policies rarely reach it.

5.2 Different routes embody different styles of inquiry

Table 2. Navigation regimes encode different epistemic priorities.

Policy	Institutional strength	Characteristic risk
Semantic	Fast direct relevance	Similarity trap; weak relational reasoning.
Local	Explainable neighborhood detail	Seed dependence and local confinement.
Diffusion	Associative integration and multi-hop reach	Hub amplification and centrality bias.
Community	Corpus-wide coverage	Summary loss and community-boundary errors.
Typed path	Domain logic and auditability	Ontology dependence; missed untyped links.
Temporal	Decision lineage and state reconstruction	Historical lock-in or obsolete precedent.
Contrastive	Dissent and disconfirmation	Cost and false equivalence.
Exploratory	Weak ties and cross-community discovery	Novelty flood and reviewer burden.
Hybrid	Risk-sensitive policy composition	Opaque weighting if the plan is not recorded.

5.3 The recursive effect: a worked NB15 fork

NB15 makes the abstract claim numerical. It starts nine policies from the same 48-node, 55-edge governed vault. It freezes the six-question cycle, deterministic answer interpreter, context budget of ten, seed, and five governance gates. Each fork then runs forty reasoning-and-write-back episodes.

Consider two of those otherwise identical forks:

Policy after 40 episodes	Nodes	Edges	Added edges	Approved objects
Semantic	88	228	173	40
Contrastive	88	255	200	40

Both forks admit forty new insight objects, yet the contrastive route creates 27 more typed edges—15.6% more policy-conditioned additions—because it retains balanced adverse and supporting evidence more often. Under NB15’s policy-labeled edge signature, the pairwise 1–Jaccard distance is 1.00: the policies have created disjoint sets of named additions. That strict distance partly reflects policy-specific object identifiers, so the edge-count contrast is the more intuitive classroom statistic. The lesson is not that contrastive navigation is better. It is that identical initial memory and identical gates can become structurally different memory solely because the route changed what was proposed.

5.4 Contradiction as an institutional asset

In finance, contradiction is often the most valuable retrieved relation. A system that optimizes for coherence can erase dissent precisely when dissent is decision-relevant. Contrastive navigation asks whether the leading view survives a structured challenge and records what opposing branch was sought, found, or absent.

5.5 Explainable access

Source citation is not navigation provenance. A navigation record should include graph, policy, and query versions; seeds and permitted edge types; visited and retained nodes; rejected branches; budget and stopping reason; access profile; context hash; model; candidate edits; and governance decisions. This reveals not only what the model cited, but how the evidence set became possible.

5.6 Plural navigation as checks and balances

Consequential decisions should use distinguishable branches: direct relevance, typed structure, history, dissent, and exploration. The integrator reports agreement, disagreement, omissions, and sensitivity. If evidence is insufficient, the correct result is abstention, a larger budget, external search, or human escalation.

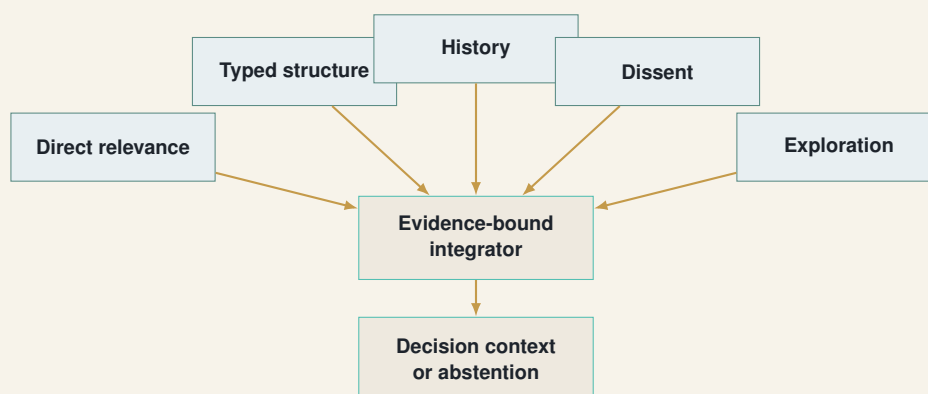


Figure 6. Plural navigation constructs separate evidence branches before an integrator reports agreement, conflict, omission, and sensitivity.

5.7 From paths to a context portfolio

Plural paths still exceed the finite context budget. Selection is therefore a portfolio problem in the literal language of finance. Candidate paths are assets. Paths that repeat the same source, assumption, or community are correlated exposures rather than diversification. The dissent branch is a hedge: it may reduce apparent coherence while protecting the committee against a crowded evidentiary position. The

context budget is capital, and every token allocated to one branch carries an opportunity cost. Selection is therefore a mean-variance problem under a coverage constraint: maximize expected decision-relevant information, penalize covariance and provenance concentration, require minimum exposure to support, mechanism, precedent, dissent, and weak signals, and remain inside permission and token budgets.

Unlike a traded portfolio, path returns are not stationary and covariance is partly semantic, institutional, and genealogical. Two differently worded paths can be perfectly correlated if they descend from the same source. A high-relevance portfolio can still be fragile if it excludes the one low-probability branch that identifies a covenant, legal prohibition, or regime break. For a finance audience, the right intuition is not “retrieve the five best passages” but “allocate scarce cognitive capital across non-independent evidence exposures while preserving a hedge.”

Typed search, submodular selection, integer programming, simulated annealing, learning-based routing, and quantum-conditioned sampling can all sit behind the same admissible-path and packing interface. No method is privileged by its metaphor. Every one must climb the numbered claim ladder in Teaching Box 1: difference, usefulness, and only then demonstrated advantage.

The quantum notebooks are therefore an extension of the knowledge pipeline, not its culmination and not a separate claim about institutional intelligence. **NB18** asks whether quantum-walk dynamics create a measurably different distribution over admissible paths. **NB19** asks whether a QUBO/QAOA selection formulation improves the packed portfolio under the same evidence, coverage, and resource constraints. Both sit beside strong classical comparators. If a classical heuristic dominates, the governed architecture remains intact and the correct scientific result is to retain the classical method. The reusable contribution is the common selection contract and claim discipline, not a presumption of quantum advantage.

5.8 Access-conditioned navigation

In finance, the admissible graph is not common to all users. Material non-public information, separation between public and private sides, wall-crossing, restricted lists, client confidentiality, research independence, need-to-know controls, and jurisdictional limits constrain which nodes and edges a user may traverse. Access is not merely a filter applied after retrieval; it changes the set of admissible paths, the available counter-evidence, and the stopping rule.

For user u , write:

$$C_{t,u} = N(G_t | \mathcal{A}_{t,u}, q_t; \pi_t, B_t, \Omega_{t,u}),$$

where $\mathcal{A}_{t,u}$ is the access-permitted subgraph. There is therefore no single cognitive topology for the institution. Two analysts asking the same question can legitimately receive different contexts because one is wall-crossed and the other is not.

(a) Reconstruct this user’s admissible world. The **navigation record** (Vocabulary Card, p. 9) already carries `access_profile`. Audit must therefore reconstruct what *this user* could have seen at the time, not what an omniscient auditor can see today. The structural consequences are significant. The knowledge-contribution measure becomes $KC_u(k)$, because an object may be valuable to one permitted population and invisible to another. The topology audits in §10.5 also become user- or role-relative: contradiction reachability, peripheral-node survival, community coverage, and authority concentration must be evaluated over the subgraphs users are actually allowed to see. Audit access must not be mistaken for trading or advisory permission.

(b) Existence can itself be MNPI. Redaction cannot operate only on node contents. The existence of a node, relationship, restricted project, counterparty cluster, or unusual retrieval frequency may itself

disclose a transaction, mandate, investigation, or position. An access-conditioned graph must therefore protect identifiers, degree, adjacency, counts, and negative responses where those metadata reveal the hidden object.

(c) Write-back is a wall-crossing vector. An edge induced on the private side that later surfaces to a public-side navigator is a compliance incident even if neither endpoint's text is reproduced. The governance operator Γ_t cannot check only the candidate's content, provenance, and reviewer. It must evaluate the access-lattice implications of the *relationship itself*: endpoint classes, permitted inheritance, cross-domain reachability, derived sensitivity, and the audiences to whom the new edge would become traversable. Approval without this relation-level check is not safe admission.

Navigation principle

Govern not only what memory contains, but what each authorized system and user can systematically retrieve, what they cannot see, and how today's permitted route becomes part of tomorrow's map.

6 Governance in Operation

Classroom pairing. **NB01–NB03** make the candidate/authority boundary executable; **NB16** records and governs navigation; **NB20** separates evidence-bound roles; and **NB22** demonstrates dependency-aware reversal. The subsections on discovery, erasure, reviewer capacity, depreciation, and model-risk alignment must still be taught in prose because a successful Colab run cannot establish that an institution can legally, economically, or organizationally operate the design.

6.1 Two coupled lanes

The architecture separates a knowledge lane from an action lane. The knowledge lane moves from candidate creation through eligibility and human authority to a controlled proposal. The action lane begins only after a separately authorized execution decision. No graph edge should execute a transaction; no persuasive memorandum should grant itself approval.

6.2 Integrity, determinism, traceability, and auditability

Integrity, determinism, traceability, and auditability are defined on the Vocabulary Card (p. 9). Integrity asks whether content changed unexpectedly. Determinism asks whether the same declared inputs produce the same registry or fingerprint. Traceability asks whether an object can be followed to sources, transformations, model or rule versions, policy, and reviewer. Auditability asks whether an independent party can reconstruct the state and the decision. NB07 and NB10–NB11 turn these words into executable acceptance tests.

6.3 Layered validation and orphan control

A successful run is not a validated knowledge state. Validation is layered: *syntactic* checks enforce the output contract; *referential* checks resolve every identifier and edge endpoint; *semantic* checks enforce the domain ontology; *evidential* checks test whether cited material supports the claim; *quantitative* checks reproduce derived values from visible fields and formulas; *temporal* checks preserve validity intervals and supersession; and *governance* checks verify that the exact state change was authorized.

These layers make orphan control measurable. The system should report unregistered endpoints, unresolved entity aliases, unsupported claims, isolated objects, missing back-links, broken temporal chains, and decisions without authority. Structural integrity is necessary but insufficient: a well-formed edge can still be false, a correct calculation can still be misused, and a fluent rationale can still cite irrelevant evidence.

Division of cognitive labour

The model extracts, compares, and explains. Deterministic code calculates, resolves identifiers, enforces schemas and permissions, and writes the audit record. The graph stores relationships and temporal history. Human-readable vault notes expose a projection to analysts. Governance alone determines whether generated knowledge remains a candidate or becomes institutionally adopted.

6.4 Risk-tiered autonomy

Autonomy should be proportional to consequence and reversibility. Low-risk classification can be automated with exception queues. Research navigation can be autonomous while write-back remains proposal-only. An investment recommendation should require evidence plurality, dissent search, and

explicit authority. Capital movement, client communication, and public disclosure need separate execution controls.

What the notebooks cannot show you

The next three lessons transfer furthest because a clean synthetic run cannot render their institutional weight. They concern legal exposure, scarce human attention, and the slow cost of forgetting.

6.5 The audit record is also a discovery liability

A complete record supports reproducibility and oversight, but may also become discoverable, sensitive, or legally consequential. Institutions must define retention, privilege, segregation, access, legal hold, and deletion policy before generating exhaustive reasoning traces. “Record everything” is not a governance strategy. The right record is sufficient for accountability, scoped to purpose, protected by access, and retained for a declared period.

Mandated erasure creates a direct collision with no-silent-overwrite. Under GDPR or another binding deletion obligation, an institution may be required to erase a source even though reproducibility would prefer to preserve it. The harder question is dependency: what happens to summaries, embeddings, relationships, recommendations, and decisions derived from the erased source? A production design needs an erasure ledger, legal basis, dependency traversal, cryptographic or physical deletion evidence, and a policy for whether derived objects are erased, quarantined, recomputed, or retained in irreversibly de-identified form. Treating erasure as a storage API call is one of the ways projects stop [11].

6.6 Reviewer capacity is the binding constraint

Governance is a queueing system. If candidate arrivals exceed reviewer capacity, review ages, approval quality drifts, exceptions accumulate, and informal bypasses appear. Deterministic checks should absorb schema validation, identity, permissions, hashes, endpoint integrity, and policy compliance. Human attention should concentrate on epistemic judgment, conflicting authority, material consequences, and exceptions.

6.7 Knowledge depreciation: a mechanism, not a symbol

The loss term L_t is implemented by lifecycle rules, and $\ominus$ means demotion from active to archival state, never removal. The **domain knowledge steward** owns the depreciation queue and the dependency analysis; source owners supply evidence, risk/legal/compliance may impose immediate restriction, and the governance committee funds the service level.

Exactly five operational triggers populate L_t :

1. expiry of `valid_to`;
2. supersession by a newly approved object;
3. source withdrawal, cascading to every dependent object;
4. a regime-conditioned claim whose declared regime has ended; and
5. demotion of objects whose measured contribution satisfies $KC(k) \approx 0$ across the approved held-out tasks and access profiles.

Each trigger produces one of four actions: reaffirm, narrow scope, demote to archive, or quarantine pending review. Mandatory erasure is handled under the separate legal mechanism above; it is not disguised as ordinary depreciation.

The institution must pre-allocate a maintenance budget by risk tier. A practical pilot might reserve at least 25% of reviewer capacity for maintenance and require high-risk objects to be reviewed within ten business days of a material trigger, medium-risk objects within thirty, and low-risk objects in a scheduled batch. Those numbers are policy choices, not universal constants; what matters is that they are declared and measured.

The harder capacity concession

If review capacity binds admission, it binds pruning harder. Admission arrives with a visible project and sponsor; depreciation arrives later, produces no celebratory demo, and requires understanding the downstream dependencies of an object. A system that funds ingestion but not retirement will accumulate epistemic debt even when every original admission was defensible.

6.8 Mapping to model-risk management

The architecture does not need a novel governance bureaucracy before it can enter a regulated bank. It can be mapped onto existing model-risk management (MRM) disciplines. Risk tiers approximate model materiality or model tiering; the registry is an extension of the model inventory to prompts, policies, knowledge versions, and derived objects; independent replay and outcome analysis extend validation and ongoing monitoring; and the dissent branch operationalizes *effective challenge*.

Architecture	MRM analogue	Approval implication
Risk tier	Model materiality/tier and proportionate control	Review, validation, monitoring, and change control scale with consequence.
Registry	Model inventory	Inventory the complete decision system, not only the foundation model endpoint.
Navigation and write-back records	Development/use documentation and change log	Reconstruct intended use, evidence state, overrides, and mutations.
Dissent branch	Effective challenge	Give objective, informed reviewers the standing, evidence, and authority to force change.
Replay, ablation, outcome analysis	Validation and ongoing monitoring	Test conceptual soundness, limitations, drift, and realized performance.

Historically this mapping was expressed through SR 11-7 in the United States. As of April 2026, SR 26-2 supersedes SR 11-7 and retains a risk-based approach, model inventory, validation, ongoing monitoring, and effective challenge [9]. The PRA's updated SS1/23 similarly requires an MRM framework across the model lifecycle with intensity proportionate to model tier [10]. The mapping is not a claim that every language-model workflow falls automatically inside the legal definition of a model; it is a pragmatic route by which a graduate can connect the architecture to an approved control function rather than present it as a parallel regime.

6.9 Governance creates option value

Governance is not only friction. Versioning, provenance, quarantine, and reversibility preserve the option to compare states, roll back contamination, change models, reinterpret old decisions, and expand

autonomy later. These options are valuable precisely because institutional decisions are made under uncertainty and some errors are difficult to reverse. **NB22** makes that promise executable: approve one wrong edge, enumerate its complete downstream dependency cone, reverse every dependent decision and derived relation, and prove that the restored canonical graph contains no surviving dependency on the poisoned edge.

7 The Economics of a Low-Cost, Modular, Model-Agnostic Architecture

7.1 The adoption objective

Mass adoption cannot depend on bespoke retraining for every team, unlimited context, or an architecture rebuilt whenever the preferred model changes. The fixed cost should create reusable institutional assets while marginal cost per new use case declines. Durable learning belongs primarily in governed external memory; reasoning capacity remains competitively sourced.

7.2 Modularity

Separate source ingestion, canonical representation, embeddings and indexes, graph traversal, query compilation, inference, orchestration, evaluation, governance, and external action. Explicit contracts isolate failure. A vector index can change without changing canonical identities; a new policy can be tested behind the same context interface; a risk rule can quarantine write-back without disabling read-only research.

7.3 Scalability through reusable objects

The scalable unit is the canonical object and governed transformation, not the prompt template. Once a claim, decision, or outcome has stable identity and provenance, research, investment, compliance, post-mortem, and future-model workflows can reuse it. One investment in knowledge quality creates economies of scope.

7.4 The cost curve

Let F be the fixed cost of corpus design, schemas, governance, and evaluation; let c_m, c_r, c_g denote model, retrieval, and governance costs. For n governed episodes:

$$\bar{C}(n) = \frac{F}{n} + c_m + c_r + c_g.$$

Value appears when fixed assets are reused and governance becomes risk-sensitive rather than universal. Cheap inference is not sufficient: a system that pollutes memory creates deferred correction, audit, switching, and incident costs.

Illustrative magnitudes – July 2026

Consider one non-trivial decision episode with 100,000 input tokens and 10,000 output tokens on a mid-priced frontier model. At illustrative published prices of \$2.50 per million input tokens and \$15 per million output tokens, $c_m \approx \$0.40$. Let retrieval and storage add $c_r \approx \$0.10$. Only three reviewer minutes at a fully loaded \$120 per hour imply $c_g \approx \$6.00$. Governance is therefore about fifteen times model cost before legal, compliance, exception handling, or depreciation work. The figures are illustrative rather than a procurement benchmark, but the order of magnitude is the lesson: at current token prices, c_g can dominate c_m by well over one order of magnitude [12].

7.5 Model agnosticism as bargaining power

Model agnosticism does not claim that models are equivalent. It means the institution can route tasks by accuracy, reasoning, latency, privacy, and price while preserving its knowledge and governance layer. A model swap is then an evaluation event, not a memory-loss event. Portability also supports confidentiality

strategy: highly restricted episodes can be routed to a more controlled deployment while lower-sensitivity work uses another provider, without fragmenting canonical identity. The routing decision must itself be policy-controlled and recorded.

8 The Architecture in Finance

8.1 Why finance is the proving ground

Finance combines changing facts, uncertain mechanisms, legal constraints, market reflexivity, confidential information, and accountable action. Recommendation quality depends on evidence and authority; yesterday's correct view can become today's dangerous precedent if its assumptions are not maintained. The public/private-side constraints in §5.8 make the architecture role-relative from the start.

8.2 A running example: the governed investment committee

Consider a proposed acquisition of a regional payments company. The protected source layer contains the current operating model, customer-concentration data, legal opinions, previous committee minutes, a covenant schedule, and the relationship history with the seller. Human intelligence frames the decision: whether the strategic option value justifies the integration and liquidity risk. Generative AI constructs comparisons, traces exposures, and drafts the base memorandum. A contrastive navigation branch then surfaces one contradiction: the model assumes customer portability, while a three-year-old legal memorandum states that two anchor contracts require consent after a change of control.

The AI recommends deferral. The committee overrides that recommendation because the deal lead has obtained oral comfort from both clients, but it does not erase the contradiction. The canonical record stores the evidence, the dissent branch, the model recommendation, the committee's override, its owner, the assumption that written consents will follow, and a review trigger. Three years later, one client has consented and the other has exited; the realized concentration and integration costs are attached to the original decision. The institution has not merely saved a memorandum. It has preserved a decision episode in which one contradiction surfaced, one accountable override was made, and one observable outcome later qualified the precedent.

8.3 The finance decision cycle

Frame the decision; construct the access-permitted context; generate and challenge; validate evidence; authorize knowledge changes; decide; separately authorize action; observe outcomes; and supersede or depreciate prior objects. Each step has a distinct record and owner.

8.4 Representative applications

Credit underwriting, M&A screening, portfolio construction, risk committees, regulatory interpretation, treasury, product approval, operational incidents, and client coverage all benefit from remembering not only the answer but the conditions, dissent, and outcome.

8.5 Separation of duties in an agentic system

Search agents can be autonomous in retrieval while non-authoritative in admission. Synthesis agents cannot approve their own evidence. Knowledge stewards do not thereby gain execution authority. Trading, communication, and public disclosure are governed by separate permissions and approvals.

8.6 The competitive advantage

The proposed advantage is cumulative, portable, and contestable. Later work inherits validated structure; a new model can consume the same memory; and users can challenge or supersede the basis of a

conclusion. These properties are hypotheses about strategic value until tested with institutional data.

9 Implementation: From Pilot to Cognitive Operating System

9.1 The governing rule: build the substrate before autonomy

The classroom notebooks instantiate the logical contracts of the architecture; they are not a production deployment recipe. An institution must add its own identity system, secrets and key management, network boundary, data-loss prevention, endpoint approval, retention and residency controls, security monitoring, service continuity, legal basis, and incident response. The value of the laboratory is that these production controls attach to explicit objects and transitions rather than to an undefined promise that an AI system is “secure.”

The implementation order is:

value and risk → sources → canonical knowledge → navigation
→ context portfolios → controlled action.

Do not read this as a waterfall requiring perfection. It is an authority rule: later autonomy may not erase the contracts established earlier.

9.2 Stage 0: define the value case and constitutional perimeter

Stage 0 has no notebook. Choose one narrow, recurring, consequential decision domain. Declare corpus boundary, access classes, decision owner, candidate and authority states, approved model/deployment pattern, retention and residency, review budget, rollback, and the evaluation rubric. Freeze the minimum improvement and reviewer-hour threshold before seeing outcomes.

9.3 Stage 1: governed sources

NB01–NB03 establish candidate authority through *Candidate Knowledge*, the *Governance Pipeline*, and *Human Approval and Provenance*. **NB04–NB07** then establish corpus discovery, metadata, the manifest, integrity, and validation. Table A.2 is the operational specification; the points below are the two production realities a clean notebook cannot absorb.

9.3.1 Entity resolution

NB07 deduplicates artifacts by hash; it does not resolve entities. A real counterparty can appear under five names, several subsidiaries, a legacy ticker, a post-merger identity, and multiple security identifiers. Production ingestion needs an entity-resolution service that preserves aliases and lineage; maps LEI, CUSIP, ISIN, ticker, and internal IDs; records effective dates for symbol changes; distinguishes legal entity from brand; represents parent-subsidiary hierarchies; and routes uncertain matches to review. A false merge contaminates every downstream edge, while a missed merge fragments precedent.

9.3.2 Corpus reality

Institutional logic does not arrive as clean text. It lives in scanned PDFs, spreadsheet formulas and hidden cells, slide speaker notes, email attachments, handwritten annotations, and versions in which the memo number disagrees with the model number. Stage 1 must preserve the original artifact, extraction confidence, page or cell locator, formula/value distinction, OCR provenance, and cross-artifact contradiction. Normalization without forensic linkage creates a tidy but fictitious corpus.

What broke: documents → canonical objects

A file system cannot answer which recommendations depended on an assumption that has since been superseded. Stable object identity, status, and lineage are forced by that failure.

9.4 Stage 2: canonical objects and graph

NB08–NB11 build canonical knowledge objects, their relationship graph, the CKO registry, and the validation-and-statistics report. Read the Canonical Vocabulary and Table 3 before the live build. The next notebook, **NB12**, deliberately changes the object of study from stored structure to question-induced structure and therefore begins Stage 3.

What broke: objects → graph

Identity without relation cannot traverse precedent. A collection of perfect objects still cannot express which evidence supports, contradicts, informs, supersedes, or resulted from which decision.

9.5 Stage 3: inquiry and navigation

NB12 first makes the question an architectural input. **NB13–NB16** then freeze the graph, model, prompt, question, budget, interpreter, and gates; vary navigation; allow governed write-back; observe the fork; and select a policy under an auditable governance record. Section 5.3 supplies the worked magnitude. The synthetic-design boundary is stated once, in Section 4.5. In production, the same stage also verifies the governed context boundary in §2.5: every retrieved node and induced relation must be eligible for the user, purpose, destination model, and intended output channel.

As the shared laboratory for *Question-Induced Knowledge Architecture* and *AI-Driven Knowledge Engineering*, **NB12** also reuses the three-vault control in §4.2, builds the corporate decision ontology, turns narrative shocks into reproducible exposures, and projects the engineered state into complementary artifacts. Markdown or Obsidian notes support human inspection; CSV registries support deterministic tests; GraphML supports traversal; and visualizations support classroom and committee communication. No projection is the sole source of truth: all must be rebuildable from the versioned registries and provenance.

What broke: graph → navigation policy

The vault exceeds any context budget. A graph can store all permitted relations, but a model still needs a reproducible rule for which subgraph becomes cognitively active.

9.6 Stage 4: context portfolios

NB17–NB19 construct admissible path encodings, compare quantum walks with classical traversal, and formulate context-portfolio selection as QAOA/QUBO beside strong classical baselines. Before an advanced selector influences production, record its frozen candidate set, classical baseline, evidence and compute budgets, noise and latency assumptions, and highest supported rung in Teaching Box 1. The NB19 result is intentionally a lesson in stopping: observed difference did not become advantage.

What broke: plural paths → portfolio selection

Five evidence branches do not fit either. Plural retrieval exposes support, dissent, history, mechanism, and weak signals; a packing rule must decide which combination earns the finite context budget.

9.7 Stage 5: deliberation, governed write-back, and reversal

NB20 conditions a multi-agent deliberation on the selected evidence portfolio and keeps roles bound to attributable evidence. **NB21** subjects governed write-back to noise and tests any claimed advantage against the strongest declared comparator. **NB22** then introduces an approved-but-wrong edge, traces its dependency cone, and proves whether the active graph can be restored. Only after sources, objects, navigation, evaluation, and rollback are stable should production agents receive bounded, reversible tools. Start with proposal-only permissions, then low-consequence action under explicit policy. Preserve the ability to abstain, escalate, interrupt, and roll back.

9.8 The minimum production stack

Table 6 is a controlled minimum, not a vendor checklist. Each layer must expose a version, owner, access rule, and acceptance test.

Table 6. Minimum production stack. Every layer exposes a version, owner, access rule, and acceptance test.

Layer	Minimum production capability
Source admission	Immutable or versioned artifacts; authority, validity, integrity, permission, and corpus manifest.
Representation	Canonical objects, typed relationships, provenance, epistemic class, governance status, supersession, and validity intervals.
Retrieval/navigation	Semantic, typed, temporal, contrastive, and escalation routes; policy registry and context record.
Reasoning	Replaceable model interface; schema-constrained output; evidence-bound claims and abstention.
Governance	Deterministic eligibility, human authority, risk tier, quarantine, queue metrics, rollback, and depreciation.
Evaluation	Frozen cases, blinded rubric, model-portability tests, ablation, review hours, topology audits, and incidents.
Action	Explicit tool permissions, separation of duties, approval, reversibility, and post-action reconciliation.

9.9 Organization and ownership

The operating model requires explicit authority and funded capacity:

- ▶ the **domain executive** owns the decision outcome, the value case, and the business P&L that funds ordinary reviewer hours;
- ▶ **source owners** attest authority, access, retention, and withdrawal;
- ▶ the **knowledge steward** owns object quality, entity resolution, supersession, archival demotion, and the depreciation queue;
- ▶ the **navigation-policy owner** owns admissible paths, context budgets, access-conditioned routing, and topology diagnostics;
- ▶ **model risk** owns tiering, inventory alignment, independent validation, limitations, and change

control;

- ▶ **legal and compliance** own privilege, MNPI, wall-crossing, restricted-list implications, retention, erasure, and incident escalation;
- ▶ the **platform operator** owns availability, identities, permissions, logging, rollback tooling, and environment fingerprints; and
- ▶ an **independent evaluator** owns the frozen cases, blinded rubric, leakage controls, statistical analysis, and publication of negative results.

Review SLA. High-risk admission or material access-lattice changes: two business days; medium-risk proposals: five; low-risk batch review: ten. Depreciation triggers use the separate ten/thirty/batch schedule in §6.7. Queue age, exception rate, reversal time, and dependency depth are reported monthly to the domain executive and quarterly to model risk.

Funding rule. The use-case P&L pays ordinary admission and maintenance because it receives the benefit. A central risk/platform budget pays independent validation, shared controls, and incident response. When reviewers span businesses, chargeback follows candidate volume weighted by risk tier and dependency depth, not token consumption. Unfunded reviewer hours are not an efficiency; they are a hidden subsidy that invalidates the business case.

A RACI chart alone is insufficient. Every lifecycle transition must name the role whose authority changes state, the SLA, the escalation path, and the budget that purchases the required human attention. These are the operating facts Colab cannot demonstrate.

10 Institutional Validation Protocol

Classroom pairing. Read this section and Appendix C before **NB21**. The notebook establishes whether the controlled mechanism executed as specified. The validation protocol asks the harder institutional question: whether governed persistence improves a real decision domain after leakage, review time, maintenance, incidents, and opportunity cost are counted.

10.1 The reader's experiment

This paper cannot estimate institutional value without an institutional corpus, decision process, and outcome record. An institution that holds those assets can test the mechanism. User counts, prompt volumes, generated words, and graph density do not establish that decisions improved or reusable knowledge accumulated.

10.2 The five-arm comparison

Within one recurring domain, run comparable held-out cases under the same eligible evidence, model class, task protocol, and budget:

1. raw admitted facts;
2. static RAG;
3. static GraphRAG;
4. question-induced ungoverned persistence;
5. question-induced governed persistence.

The fourth arm is the critical control. It may show that persistence creates most of the benefit and governance is not worth its review cost. A paper about governed memory must name and accept that possibility.

10.3 Controls, measures, and sequence

Freeze the model, evidence and context budgets, source boundary, prompt, evaluation cases, outcome window, and rubric. Randomize or prospectively match cases where possible. First test static answer quality on frozen states. Then allow write-back and test later decisions against the resulting forks. Finally, apply common shocks and counterfactual navigation policies.

Retrospective replay. Before a prospective pilot, replay historical decisions with the corpus truncated to the exact decision date. Freeze the source manifest, object versions, market data, access profile, model, prompt, and policy as they could legitimately have existed then; hide later committee minutes, realized outcomes, restatements, subsequent research, and any derived object that depends on them. Compare the replayed recommendation with the contemporaneous decision and later outcome, but never allow the later outcome into the model's context.

Leakage discipline. Time is a hard authorization boundary. Split by decision episode, not by document chunk; fit entity resolution, embeddings, community detection, policy tuning, prompt selection, thresholds, and normalization statistics only on the training period; and rebuild the complete eligible graph at each cutoff. A held-out memorandum is not held out if its conclusion survives in an edge, summary, vector index, taxonomy label, or reviewer-authored prompt.

10.4 Knowledge contribution by ablation

For object k , estimate contribution on held-out tasks:

$$KC_u(k) = P(V_{t,u}) - P(V_{t,u} \setminus \{k\}),$$

where u indexes the relevant access profile. The estimate is not additive because objects interact. Group ablations should test communities, relation types, decision histories, navigation records, and complete episodes.

Table 5. A balanced validation protocol measures present output and the state inherited by future decisions.

Dimension	Representative measures	Test question
Decision value	Blind expert ranking, calibration, realized outcome, regret, scenario response	Are decisions better rather than merely faster?
Productivity	Cycle time, analyst hours, marginal cost, reuse rate	Does the system release scarce analyst and reviewer attention?
Knowledge contribution	Held-out uplift, ablation, cross-task reuse, model-portability retention	Does accumulated memory improve later work?
Epistemic health	Provenance coverage, contradiction reachability, stale and orphan rates, authority completeness	Is memory becoming more trustworthy and plural?
Navigation health	Policy overlap, role-relative community coverage, centrality exposure, exploration yield, abstention	What does each permitted population repeatedly see or miss?
Governance economics	Review burden, queue age, approval drift, rollback, incident cost, depreciation backlog	Is control producing value net of its capacity cost?

Table 5 (Table 5) must be read with Table 4 (Table 4): a method is not evaluated only by answer quality, and the same architecture can have different cost and control profiles under different implementation strategies.

10.5 Portability and user-relative topology audits

Freeze the vault, policies, questions, and budgets; swap the model; measure retained decision performance, provenance fidelity, recommendation stability, cost, and latency. If advantage disappears, the derived layer may encode one model's rhetoric rather than portable structure.

Periodic topology audits should examine, for each relevant access profile, authority and degree concentration, retrieval exposure, contradiction reachability, peripheral-node survival, bridge precision, relation diversity, temporal coverage, and reviewer acceptance drift. Graph measures are diagnostics, not truth measures.

10.6 The claim ladder for advanced methods

Teaching Box 1 The three-rung claim ladder

1. Difference. The method's output distribution or selected context differs from the declared baseline.

2. Usefulness. Under equal evidence and resource budgets, that difference improves a task metric that matters.

3. Demonstrated advantage. The improvement survives end-to-end execution, scale, latency, noise, integration cost, and comparison with the strongest relevant classical method.

Worked result. On a 38-node, 54-edge synthetic graph, a quantum-conditioned selector produced a different graph trajectory but lower declared composite utility than the classical selector: approximately **0.793 versus 0.906** at zero injected noise. The experiment established difference and stopped at rung one. It did not establish usefulness or computational advantage.

The ladder applies equally to learned routers, evolutionary search, annealing, quantum-inspired methods, and physical quantum backends. It also applies to this paper. The controlled laboratories demonstrate that governed write-back and navigation policy can produce different contexts and different memory trajectories. The paper's own central claim therefore reaches *difference*. Whether those differences are useful in an institution, net of review and maintenance, remains open. Modelling the discipline on our own architecture is more instructive than applying it only to vendors.

10.7 Pre-registered refutation conditions

A claim is evaluable only with a common rubric and explicit constraints. The course design declares **one primary measure**, not three:

$$E = \frac{\bar{Q}_{\text{governed}} - \bar{Q}_{\text{ungoverned}}}{\text{incremental reviewer-hours per decision}}.$$

Here Q is the blinded 100-point decision-quality rubric and the denominator is fully loaded review time, including exceptions and reversals. All other quantities in Table 5 are secondary outcomes, safety constraints, or diagnostics.

For a matched replay of $n = 50$ decisions, assume for course design a within-pair standard deviation of 12 rubric points. At two-sided $\alpha = 0.05$ and 80% power, the approximate minimum detectable raw effect is

$$(1.96 + 0.84) \frac{12}{\sqrt{50}} \approx 4.8 \text{ points.}$$

The pre-registered course threshold rounds this to five points and adds a cost ceiling:

Pre-registered threshold

Governed persistence must beat ungoverned persistence by at least $X = 5$ rubric points while consuming no more than $Y = 1.5$ incremental reviewer-hours per decision. Equivalently, the observed primary efficiency must satisfy $E \geq 3.33$ rubric points per reviewer-hour, with the raw five-point uplift also achieved. Both arms must meet the same minimum provenance coverage and incident limits. A domain team may choose different X , Y , variance assumption, power, or sample size, but must justify and freeze them before assignment or outcomes are revealed.

If the threshold is not met, the central usefulness claim fails for that tested domain. It is also weakened if gains vanish under a model swap, depend on inaccessible dissent, or are financed by deteriorating provenance, topology, or depreciation backlog. The response is to stop, redesign, repeat, or reject; not to rename the metric after seeing the result.

11 Leadership Agenda and Conclusion

11.1 What leaders must decide

Leaders must decide which knowledge is genuinely differentiating, which questions deserve durable memory, which sources and users belong inside each perimeter, who can change canonical state, what review and depreciation capacity will be funded, and which actions remain outside agent authority.

11.2 Ten commitments for institutional adoption

1. Treat the model as a component, not the system.
2. Treat proprietary knowledge as a strategic asset and disclose only the minimum eligible context, not the vault itself.
3. Preserve source identity and authority.
4. Keep candidates distinct from authoritative knowledge.
5. Represent decisions, assumptions, dissent, and outcomes.
6. Govern deployment, context construction, navigation, access, and write-back, not only storage.
7. Build reviewer and depreciation capacity into the operating model.
8. Preserve reversibility and model portability.
9. Pre-register thresholds and retain negative results.
10. Scale only what can be evaluated, audited, maintained, and reversed.

11.3 The close: intellectual transition, economics, and responsibility

The transition runs from answers to governed episodes, from documents to addressable knowledge, from stored graph to cognitive topology, and from one-period output to inherited memory. Its business case is not simply labor substitution: a governed episode may lower future search cost, preserve rare rationale, and expose contradictions, but it may also create maintenance, discovery, access, and recursive-error costs. Institutional value is therefore an empirical result, not an architectural entitlement. Human judgment remains the source of purpose and authority; the architecture makes its evidence, conditions, and downstream consequences reconstructible.

The pipeline is one system: protected artifacts become canonical objects and a traversable graph; a governed boundary constructs the context a model may see; the model returns candidates; approved write-back changes future memory; and navigation and portfolio policies determine which evidence becomes cognitively active. Quantum methods enter only at the selection layer, beside classical alternatives and under the same claim discipline.

For the MFin cohort, the paper explains the architecture and its limits; the author-led Colab sequence lets students construct, break, compare, govern, and reverse it. Neither component alone completes the educational objective.

The final proposition

The winning institutions will not be those that merely possess the strongest model. They will be those that combine human intelligence, institutional human knowledge, and generative AI through a protected, governed interface to produce an evolving institutional memory—a cumulative system capable of searching widely, remembering carefully, acting responsibly, preserving proprietary authority, and changing its mind with evidence.

A Course sequence, sessions, and notebook ladder

A.1 Paper-section-to-session map

Table A.1. Reading-to-session map. Students complete the listed reading before the live notebook session.

Reading before class	Live notebook session	Purpose of the pairing
Vocabulary Card; Canonical Vocabulary; §6	NB01–NB03	Candidate creation, machine eligibility, lifecycle states, human authority, and rejection without canonical mutation.
§9.3	NB04–NB07	Corpus admission, metadata, manifest, integrity, entity-resolution limits, and messy-document reality.
Canonical Vocabulary; Appendix B	NB08–NB11	CKO construction, typed relationship graph, registry, validation, and statistics.
§4; Vocabulary Card; <i>AI-Driven Knowledge Engineering</i>	NB12	Questions as architectural inputs; three-vault preservation; corporate ontology; deterministic shock features; governed decisions, supersession, and projection of question-induced conditional knowledge.
§2.5; §5.1–§5.6	NB13–NB16	Navigation-policy laboratory, counterfactual context and answer comparison, recursive write-back, topological divergence, audit, and policy selection.
§5.7; §10.6	NB17–NB19	Paths as evidence objects, portfolio selection, classical baselines, quantum-conditioned distributions, and the difference/usefulness/advantage ladder without presuming quantum advantage.
§8; §9.7–§9.9	NB20	Quantum-conditioned, evidence-bound multi-agent deliberation, synthesis, separation of duties, production stack, ownership, and funded review.
§10; Appendix C	NB21	Governed write-back under noise, advantage benchmarking, primary measure, leakage discipline, MDE, and pre-registered decision rule.
Section 6.9; Appendix B	NB22	Poisoned-edge dependency tracing, complete reversal, restored-state proof, and incident post-mortem.

A.2 Full NB01–NB22 ladder

Table A.2. The canonical directory ladder: exact notebook filename, concept, governed output, and the failure produced by skipping the rung.

NB	Canonical directory filename	Concept taught	Governed output/test	What breaks if this rung is skipped
NB01	NB01_Candidate_Knowledge	Generation is not admission	Candidate objects with identity and provenance	Model prose enters memory without a distinct non-authoritative state.
NB02	NB02_Governance_Pipeline	Candidate vs. authoritative	Eligibility, quarantine, and reason-coded queues	Schema, permissions, and basic policy failures consume scarce reviewer attention.
NB03	NB03_Human_Approval_and_Provenance	Candidate vs. authoritative	Decisions, rationale, and write-back proposals	Machine validation is mistaken for institutional approval.
NB04	NB04_Corpus_Discovery	Inventory before interpretation	Reproducible artifact inventory	Sources disappear, duplicate, or enter without a declared boundary.
NB05	NB05_Metadata_Extraction	Location is not identity	Normalized descriptions and locators	Extracted claims cannot return to a precise page, sheet, cell, or note.
NB06	NB06_Corpus_Manifest	Authority must be declared	Authoritative manifest with stable IDs	Mere presence in storage is treated as permission and authority.
NB07	NB07_Integrity_and_Validation	Same input, same fingerprint	Hashes, duplicates, and exception reports	Silent artifact drift defeats reconstruction; entity resolution remains a separate production duty.
NB08	NB08_CKO_Builder	Documents become addressable claims	Canonical knowledge objects	Decisions and assumptions remain trapped inside files.
NB09	NB09_Relationship_Graph	Identity without relation cannot traverse	Typed, attributable graph	Precedent, contradiction, support, and dependency cannot be followed.
NB10	NB10_CKO_Registry	A registry is a contract	Registry and fingerprints	Object identity and lifecycle vary from run to run.
NB11	NB11_CKO_Validation_and_Statistics	Structure must fail informatively	Endpoint, provenance, identity, and statistical tests	Broken edges, orphan objects, and implausible graph structure look like knowledge.
NB12	NB12_Question_Induced_Knowledge_Architecture	Questions induce conditional structure	Preserved source/security/destination states; question records; corporate decision objects; conditioned claims; shock relations; supersession	Contingent relevance appears universal; narrative claims remain non-computable; decisions and later revisions cannot be traced as governed state transitions.
NB13	NB13_Graph_Navigation_Policy_Laboratory	Retrieval is a policy	Common policy interface	Routes cannot be compared under the same contract.
NB14	NB14_Counterfactual_Context_and_Answer_Comparison	Same graph, different context	Frozen-state context and answer divergence	Navigation effects are confounded with graph, model, prompt, question, or budget changes.
NB15	NB15_Recursive_Write_Back_and_Topological_Divergence	Today's route becomes tomorrow's map	Policy-specific graph forks	Write-back's topological path dependence remains an assertion.
NB16	NB16_Navigation_Governance_Audit_and_Policy_Selection	The route needs a record	Policy registry, risk tier, audit bundle, abstention	A result can cite sources while its evidence-construction process remains irreproducible.
NB17	NB17_Quantum_Brain_Architecture_and_Path_Encoding	Paths are admissible evidence objects	Path catalogue, baselines, encoding manifest	Portfolio methods optimize an undefined candidate space.
NB18	NB18_Quantum_Walks_Through_a_Knowledge_Graph	A different distribution is rung one	Classical and quantum distributions, sweep, noise stress	Novelty is mistaken for usefulness.
NB19	NB19_QAOA_and_QUBO_Context_Portfolios	Difference is not advantage	Exact optimum, greedy, annealing, statevector QAOA	An advanced method escapes comparison with the strongest classical baseline.
NB20	NB20_Quantum_Conditioned_Multi_Agent_Deliberation	Roles need evidence boundaries	Six roles, synthesis contract, decision memorandum	Agent plurality becomes repeated rhetoric or manufactured consensus.

NB	Canonical directory filename	Concept taught	Governed output/test	What breaks if this rung is skipped
NB21	NB21_Governed_Write_Back_Noise_and_Advantage_Benchmarking	Claims stop where evidence stops	Noisy write-back runs, utility, governance, and resource accounting	A successful build is reported as advantage without an equal-budget, evaluable claim.
NB22	NB22_Poisoned_Edge_Dependency_Reversal	Reversibility means dependency repair	Poisoned approved edge, dependency cone, reversal ledger, restored-state test	Rollback removes one edge while dependent decisions and induced relations continue to contaminate canonical memory.

A.3 One cohort, one order

The course sequence is constitutional before computational:

NB01–NB03 → NB04–NB07 → NB08–NB11 → NB12–NB16 → NB17–NB19 → NB20–NB22.

Checkpoint 1—authority. After **NB03**, students can distinguish candidate creation, machine eligibility, and human authority.

Checkpoint 2—corpus. After **NB07**, students can reproduce the admitted corpus and identify both integrity exceptions and unresolved entities.

Checkpoint 3—canonical graph. After **NB11**, an independent run can rebuild and validate canonical objects, graph, and registry from frozen inputs.

Checkpoint 4—causal identification. After **NB16**, students can hold graph, model, prompt, question, budget, and governance fixed while varying navigation.

Checkpoint 5—claim discipline. After **NB21**, students can stop at the highest evidentiary rung supported by an equal-budget comparison.

Checkpoint 6—reversibility. After **NB22**, students can traverse a poisoned edge’s dependency cone, reverse every dependent state change, and prove restoration.

The notebooks are not the permanent architecture. They make its contracts visible, executable, testable, and teachable.

B Minimum Canonical Contract

B.1 Canonical knowledge object

object_id	object_type
title	content
source_artifact_id	source_locator
created_at	created_by
generation_method	model_or_rule_version
epistemic_class	governance_status
confidence	valid_from
valid_to	supersedes
superseded_by	access_profile
integrity_hash	review_trigger
next_review_at	depreciation_owner

The permitted values of epistemic_class are:

- ▶ observed;
- ▶ asserted-by-source;
- ▶ derived-by-model;
- ▶ inferred-under-assumption; and
- ▶ contested.

Confidence, governance status, and epistemic class are separate fields.

B.2 Relationship object

relationship_id	source_object_id
target_object_id	relationship_type
question_id	reasoning_episode_id
evidence_refs	assumptions
condition	confidence
epistemic_class	approval_status
reviewer	valid_from
valid_to	parent_relationship_id
superseding_relationship_id	
source_access_class	target_access_class
derived_access_class	access_lattice_check

B.3 Navigation record

navigation_record_id	query_id
graph_version	policy_id
policy_parameters	risk_tier
access_profile	seed_nodes
allowed_edge_types	visited_nodes
retained_nodes	rejected_branches
budget	stopping_reason
context_hash	model_id

candidate_edit_ids governance_decisions

B.4 Reversal record

reversal_id	poisoned_object_or_edge_id
detection_time	detected_by
root_cause	original_approval_record
dependency_query_version	affected_object_ids
affected_decision_ids	affected_navigation_records
reversal_actions	superseding_objects
archived_objects	external_actions_reconciled
restored_graph_hash	residual_dependency_count
independent_verifier	closure_time

B.5 The canonical discipline

The schema does not claim that every field is objective. Confidence, epistemic class, and source authority can be contested. The contract makes the contest visible. A rejected candidate remains visible with a reason code; a superseded object remains available for reconstruction; and a change in model, rule, source, policy, access, or authority appears as a change in state.

Minimum safe recursion

No generated object may influence authoritative future memory unless its source, question, transformation, epistemic status, navigation provenance, access profile, and authorization are recoverable.

C Graded Assignment: Pre-register an Institutional Test

C.1 Brief

Each team receives one decision domain: for example credit renewal, M&A screening, liquidity escalation, research recommendation review, model-risk exception, or operational-incident response. The team is not asked to build a universal assistant. It must design one test whose result could make it reject the paper's usefulness claim for that domain.

C.2 Required submission

1. **Frozen design (25%).** Define eligible corpus, access profiles, case population, model class, prompt/task protocol, evidence and context budgets, outcome window, blinded 100-point rubric, randomization or matching, exclusions, and leakage controls. Include a retrospective replay with every corpus version truncated to the historical decision date.
2. **Two arms (20%).** Specify ungoverned persistence and governed persistence. Explain exact write-back, review, supersession, quarantine, reversal, and depreciation differences. Keep all other conditions equal.
3. **Pre-registered primary measure and threshold (30%).** Use one primary measure: blinded rubric-point uplift per incremental reviewer-hour per decision. Choose X , the minimum raw rubric-point improvement, and Y , the maximum incremental reviewer-hours per decision. State the paired-difference variance assumption, sample size, α , power, and resulting minimum detectable effect. Justify and freeze all choices *before* seeing outcomes. State provenance and incident constraints that neither arm may trade away.
4. **Analysis and decision rule (15%).** Define estimation, missing data, model-swap sensitivity, access-relative topology audits, and the exact rules for stop, redesign, repeat, or scale.
5. **Governance record (10%).** Assign owners, authority, queue and maintenance capacity, rollback, legal retention, and final committee accountability.

C.3 Submission template

Table C.1. Team pre-registration template.

Domain and decision owner
Eligible case population
Frozen source boundary and access profiles
Historical decision cutoff and replay corpus
Control arm
Governed arm
One primary: rubric uplift / reviewer-hour per decision
Paired SD, sample size, α , power
Minimum detectable effect
X : minimum governed uplift
Y : maximum incremental reviewer-hours
Provenance and incident floors
Leakage prevention across documents, edges, summaries, vectors, prompts
Model-swap and topology checks
Stop / redesign / repeat / scale rule

The hard part

Choosing X and Y is not administrative. It forces the team to state how much better a decision must become to justify scarce reviewer capacity, and how much control cost the claimed value can bear. That number reveals what the claim actually means.

References

- [1] W. Brian Arthur. “Competing Technologies, Increasing Returns, and Lock-In by Historical Events.” *The Economic Journal*, 99(394):116–131, 1989.
- [2] Albert-László Barabási and Réka Albert. “Emergence of Scaling in Random Networks.” *Science*, 286(5439):509–512, 1999.
- [3] Jay Barney. “Firm Resources and Sustained Competitive Advantage.” *Journal of Management*, 17(1):99–120, 1991.
- [4] Darren Edge, Ha Trinh, Newman Cheng, et al. “From Local to Global: A Graph RAG Approach to Query-Focused Summarization.” arXiv:2404.16130, 2024.
- [5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” *Advances in Neural Information Processing Systems*, 33, 2020.
- [6] James G. March. “Exploration and Exploitation in Organizational Learning.” *Organization Science*, 2(1):71–87, 1991.
- [7] Luc Moreau, Paolo Missier, et al. *PROV-DM: The PROV Data Model*. W3C Recommendation, 2013.
- [8] National Institute of Standards and Technology. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1, 2023.
- [9] Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation, and Office of the Comptroller of the Currency. *Supervisory Guidance on Model Risk Management*, SR 26-2, 17 April 2026. Supersedes SR 11-7. <https://www.federalreserve.gov/supervisionreg/srletters/SR2602.htm>.
- [10] Prudential Regulation Authority. *Model Risk Management Principles for Banks*, Supervisory Statement SS1/23, updated 16 April 2026. <https://www.bankofengland.co.uk/prudential-regulation/publication/2023/may/model-risk-management-principles-for-banks-ss>.
- [11] European Parliament and Council. Regulation (EU) 2016/679, General Data Protection Regulation, Article 17, “Right to erasure,” 27 April 2016. <https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>.
- [12] OpenAI. “API Pricing,” accessed 28 July 2026. <https://developers.openai.com/api/docs/pricing>.
- [13] Richard R. Nelson and Sidney G. Winter. *An Evolutionary Theory of Economic Change*. Harvard University Press, 1982.
- [14] Ikujiro Nonaka. “A Dynamic Theory of Organizational Knowledge Creation.” *Organization Science*, 5(1):14–37, 1994.
- [15] Alejandro Reynoso. *The Governed ExoBrain: Governance-First Cognitive Operating System Architecture*. Companion monograph, 2026.
- [16] Alejandro Reynoso. *Navigation-Conditioned Knowledge Evolution*. Companion monograph and notebook programme, 2026.
- [17] Alejandro Reynoso. *The Living Encyclopedia: Question-Induced Knowledge Architecture*. Companion monograph and computational laboratory, 2026.
- [18] Alejandro Reynoso. *The Quantum Brain: Quantum Navigation, Deliberative Intelligence, and Governed Institutional Memory*. Companion monograph and five-notebook laboratory, 2026.
- [19] Alejandro Reynoso. *AI-Driven Knowledge Engineering: From Document Vaults to Governed, Evolving Corporate Intelligence*. Pedagogical companion paper to **NB12**, Cambridge Judge Business School, 2026.
- [20] Herbert A. Simon. “A Behavioral Model of Rational Choice.” *The Quarterly Journal of Economics*, 69(1):99–118, 1955.

- [21] Ewin Tang. “A Quantum-Inspired Classical Algorithm for Recommendation Systems.” *Proceedings of STOC*, 2019.
- [22] David J. Teece, Gary Pisano, and Amy Shuen. “Dynamic Capabilities and Strategic Management.” *Strategic Management Journal*, 18(7):509–533, 1997.
- [23] Michael Wooldridge. *An Introduction to MultiAgent Systems*. Wiley, 2nd edition, 2009.